



Unit 4 · Outcome 2 · Energy Sources

Non-renewable energy sources

KK01

VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.



What “non-renewable” actually means

01 What “non-renewable” actually means definition A non-renewable energy source is one that exists in a finite stock and is replenished by natural processes far more slowly than humans consume it — effectively not at all on a human timescale. Once you burn this year’s coal, the Earth will not make more for tens of millions of years. The contrast is with renewable sources (sun, wind, flowing water, tides, geothermal heat, biomass), which are replenished on a human timescale — the sun rises again tomorrow. The dividing line isn’t “will it run out one day”

Deep time: how fossil fuels formed

02 Deep time: how fossil fuels formed 3D · scrub the timeline The four fossil fuels are, quite literally, ancient sunshine . Hundreds of millions of years ago, photosynthesis locked solar energy into the chemical bonds of living things. When that organic matter was buried before it could fully decay, then cooked under heat and pressure for geological ages, the carbon-rich remains became coal, oil and gas. Burning them today releases that stored sunlight — and the stored carbon. Show formation of: Coal (swamp plants) Oil & gas (marine plankton) ► Bury & c

The five sources, one at a time

03 The five sources, one at a time click a card to open Tap each source to expand its profile — what it is, how it’s formed or sourced, how it generates energy, and its headline advantage and disadvantage. The little FOSSIL / NON-FOSSIL flag on each card is the distinction examiners love. FOSSIL Coal solid · buried plants FOSSIL Oil liquid · crude petroleum FOSSIL Natural gas gas · methane FOSSIL Coal seam gas gas · from coal seams NON-FOSSIL Nuclear uranium · fission Coal FOSSIL FUEL · SOLID Australia’s single biggest electricity source — and the one in

The line students blur

04 The line students blur drag to sort Two confusions sink marks every year: (1) treating “non-renewable” and “fossil fuel” as the same thing (nuclear breaks that), and (2) thinking coal seam gas is a renewable or a totally different fuel (it’s the same methane as natural gas, just unconventional). Sort these chips into the right bin to lock it in. Coal Oil Natural gas Coal seam gas Nuclear (uranium) Solar Wind Hydro Geothermal Biomass Non-renewable — fossil fuel finite + formed from buried organic matter + burned ☼ Non-renewable — not a fossil fuel fi

From fuel to the power point

05 From fuel to the power point 3D · watch the chain Here’s the thread that connects four of the five: coal, gas, CSG and even nuclear ultimately just make heat to boil water and spin a turbine. The fuel differs; the back end is almost identical. That’s why this dot point sets up the energy-conversion and efficiency work in KK05–KK06. Thermal (coal / nuclear) Gas turbine ► Run the chain Thermal route: chemical (or nuclear) energy → thermal energy (boiler) → kinetic energy (steam spins turbine) → electrical energy (generator). Each arrow is an energy

Advantages & disadvantages — the exam gold

06 Advantages & disadvantages — the exam gold compare table VCAA almost always asks you to compare sources or justify a choice against criteria. Learn this grid as a mental scaffold: pick two or three rows that matter for the question and contrast them. Don't list everything — select the relevant criteria. Criterion Coal Oil Natural gas Coal seam gas Nuclear Renewable? No No No No No Fossil fuel? Yes Yes Yes Yes No Main use Electricity Transport Electricity, heat, LNG Electricity, LNG Electricity Energy density Moderate High Moderate Moderate Very high O

Criterion	Coal	Oil	Natural gas	Coal seam gas	Nuclear
Renewable?	No	No	No	No	No
Fossil fuel?	Yes	Yes	Yes	Yes	No
Main use	Electricity	Transport	Electricity, heat, LNG	Electricity, LNG	Electricity
Energy density	Moderate	High	Moderate	Moderate	Very high
Operating CO ₂	Highest	High	~½ of coal	~½ of coal*	Near-zero
Dispatchable?	Yes (baseload)	Yes	Yes (fast peaker)	Yes	Yes (baseload)
Key waste/ impact	Ash, mine, air pollution	Spills, emissions	Fugitive methane	Groundwater, salt, fracking	Radioactive waste
Status in Australia	Declining	Mostly imported	Major; transitional	Major (QLD)	Prohibited

The Australian & Victorian picture (2025–26)

07 The Australian & Victorian picture (2025–26) real figures Examiners reward students who anchor general claims in real data. Here are current, defensible numbers — use them, but quote the trend as much as the exact figure. 60.5% of Australia's 2025 electricity from fossil fuels — coal 42.7%, gas 16.2%, oil 1.7%. Renewables made up the other 39.5%. ~56% of Victoria's electricity still comes from brown coal (Latrobe Valley) — the most CO₂-intensive fuel, now being phased out. 0% nuclear generation — prohibited under the EPBC Act 1999 & ARPANS Act 1998, d

Decode the command word first

◆ COMMAND WORDS

08 Decode the command word first exam craft Before writing a word, identify what the verb is asking for. The same content earns very different marks depending on whether you were told to describe or to evaluate . Identify / Name State it — no explanation. "Name two non-renewable energy sources." → Coal and natural gas. Describe Give the features/characteristics. What it is and how it works — no judgement. Distinguish State the difference between two things, ideally on the same criterion (e.g. source of CSG vs natural gas). Compare Give both similarities

Writing like a top student — the P·E·L move

P·E·L WRITING

09 Writing like a top student — the P·E·L move model answer builder For any “explain / compare / justify” item, structure each idea as P·E·L — Point, Evidence, Link . Make the claim, back it with a specific fact or figure, then link it explicitly to the question. Walk the tabs below to see a 3-mark answer assembled. P Point E Evidence L Link Full answer Question: Explain why natural gas is often described as a “transitional” fossil fuel in Australia’s electricity system. (3 marks) Point = the claim that answers the verb. Don’t bury it. ✓ Strong point Nat



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

Six traps that quietly cost marks

EXAM TRAPS

10 Six traps that quietly cost marks avoid these TRAP 01 ✗ “Nuclear is a fossil fuel.” ✓ Nuclear is non-renewable but NOT a fossil fuel — it uses fission of mined uranium, not combustion of buried organic matter. TRAP 02 ✗ “Coal is renewable because we can keep mining it.” ✓ Renewability is about the rate of natural replenishment , not whether stock remains. Coal reforms over millions of years — far slower than we use it. TRAP 03 ✗ “CSG is a different fuel from natural gas.” ✓ Same fuel (methane) — only the source differs: a reservoir (natural gas) vs ad

Practice questions

★ PRACTICE & SELF-MARK

11 Practice questions self-mark · 24 marks VCAA-style items escalating from 2 to 6 marks. Attempt each on paper first, then reveal a weak response, an annotated strong response, and the marking guide. Award yourself marks honestly — the dock bottom-right tracks your total. Q1 Defining and naming 3 marks a) Define the term non-renewable energy source . (1 mark) b) Name the five non-renewable energy sources listed in the study design. (2 marks) Reveal responses ✗ Weaker response a) An energy source that runs out. b) Coal, oil, gas, solar, nuclear. “Runs ou

One-screen cheat sheet

✦ RECAP / CHEAT SHEET

12 One-screen cheat sheet revise fast Non-renewable energy — the essentials Definition Finite stock; replenished far slower than used → not on a human timescale. Renewable = replenished in-time (sun, wind, water, geothermal, biomass). The five (study design) Oil · Coal · Natural gas · Coal seam gas · Nuclear. First four = fossil fuels; nuclear = non-fossil non-renewable. Formation / source Coal = buried plants (peat→lignite→bituminous→anthracite). Oil & gas = buried plankton → kerogen → oil (then gas with more heat). CSG = methane adsorbed in coal seams

★ Good vs bad worked answers

Study the contrast — the same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer structure — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

point Gas is better than coal. (Vague — “better” how? No criterion.)

✓ STRONG / MODEL ANSWER

evidence Burning methane releases roughly half the CO₂ per unit of electricity that brown coal does, and gas turbines can start within minutes, so they can firm variable solar and wind when output drops.

✗ WEAK ANSWER

evidence It's cleaner and pretty quick. (No figure, no mechanism — examiner can't award the mark.)

✓ STRONG / MODEL ANSWER

link This lets gas bridge the period between a coal-dominated grid and a fully renewable one — reducing emissions now without sacrificing reliability — which is why it's labelled “transitional” rather than a long-term solution.

✗ WEAK ANSWER

link So that's why people use gas. (Doesn't reconnect to “transitional”.)

✓ STRONG / MODEL ANSWER

Full P-E-L response Natural gas is called a “transitional” fossil fuel because it emits far less CO₂ than coal yet remains dispatchable. Burning methane releases roughly half the CO₂ per unit of electricity of brown coal, and fast-starting gas turbines can firm variable solar and wind within minutes. This lets gas bridge the shift from a coal-heavy grid to a renewable one — cutting emissions now without risking reliability — so it is treated as a stepping stone rather than a permanent, sustainable solution.

✗ WEAK ANSWER

er response a) An energy source that runs out. b) Coal, oil, gas, solar, nuclear. “Runs out” misses the rate idea; solar is renewable — a careless naming error costs a mark, and they've omitted coal seam gas.

✓ STRONG / MODEL ANSWER

er response a) A non-renewable energy source exists in a finite stock and is replenished by natural processes far more slowly than humans consume it, so it is effectively not replaced on a human timescale. b) Oil, coal, natural gas, coal seam gas and nuclear.

✗ WEAK ANSWER

er response a) CSG comes from coal and natural gas comes from gas. b) Good: lots of it. Bad: pollution. Circular and vague. No mention of adsorption/reservoirs or dewatering; "pollution" isn't specific to CSG.

✓ STRONG / MODEL ANSWER

er response a) Both are methane, but conventional natural gas pools freely in a porous reservoir trapped beneath impermeable rock, whereas coal seam gas is adsorbed onto coal within underground seams and must be released by pumping out groundwater to reduce pressure. b) Advantage: it taps a large domestic resource (e.g. Queensland) supporting energy supply and LNG export. Disadvantage: dewatering can draw down and contaminate groundwater and produces large volumes of saline "produced water".

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Worked good/weak answers illustrate exam technique — always check against current VCAA marking advice.